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Learning Strategy & Overfitting Prevention pada MNIST

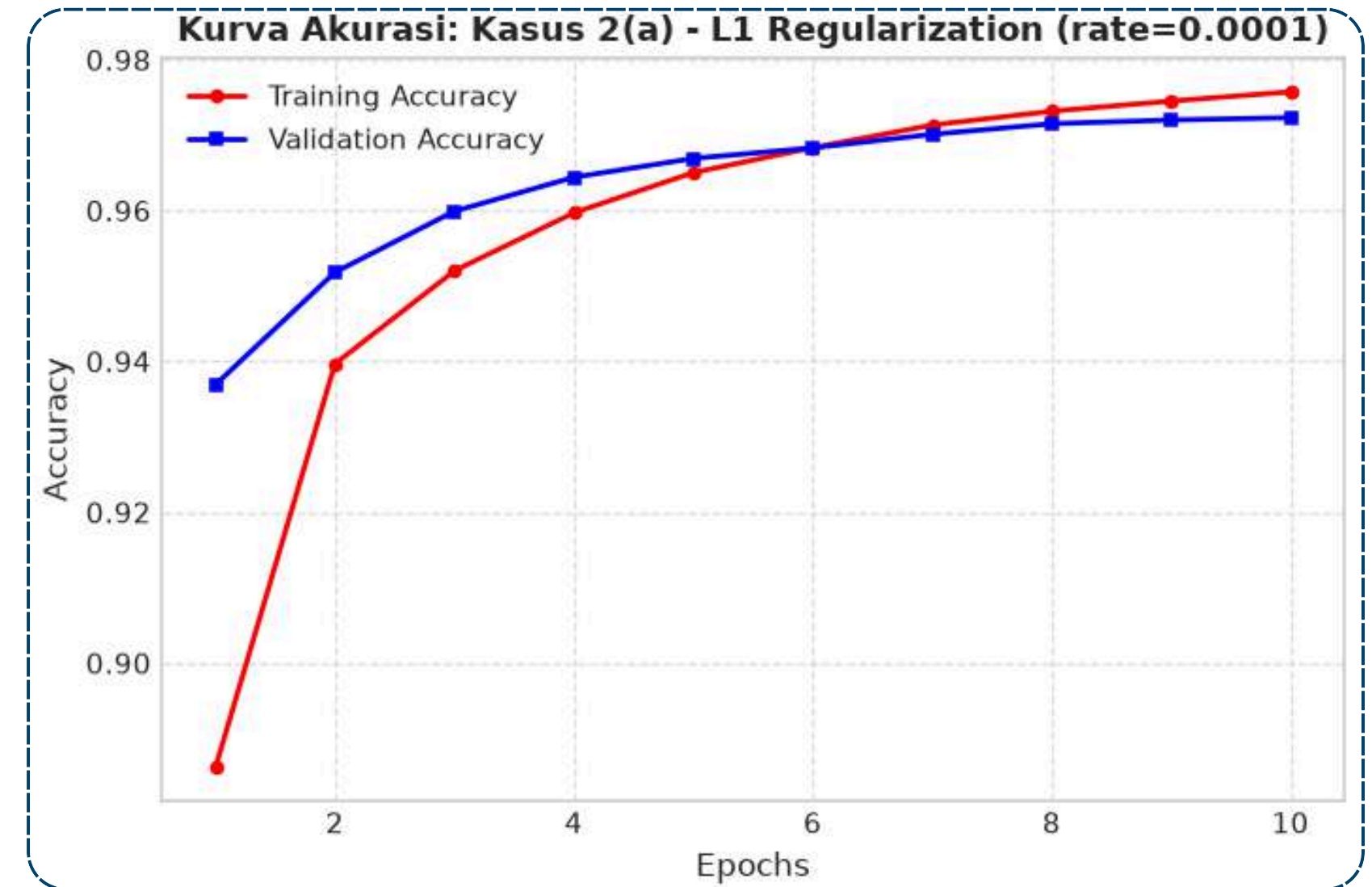
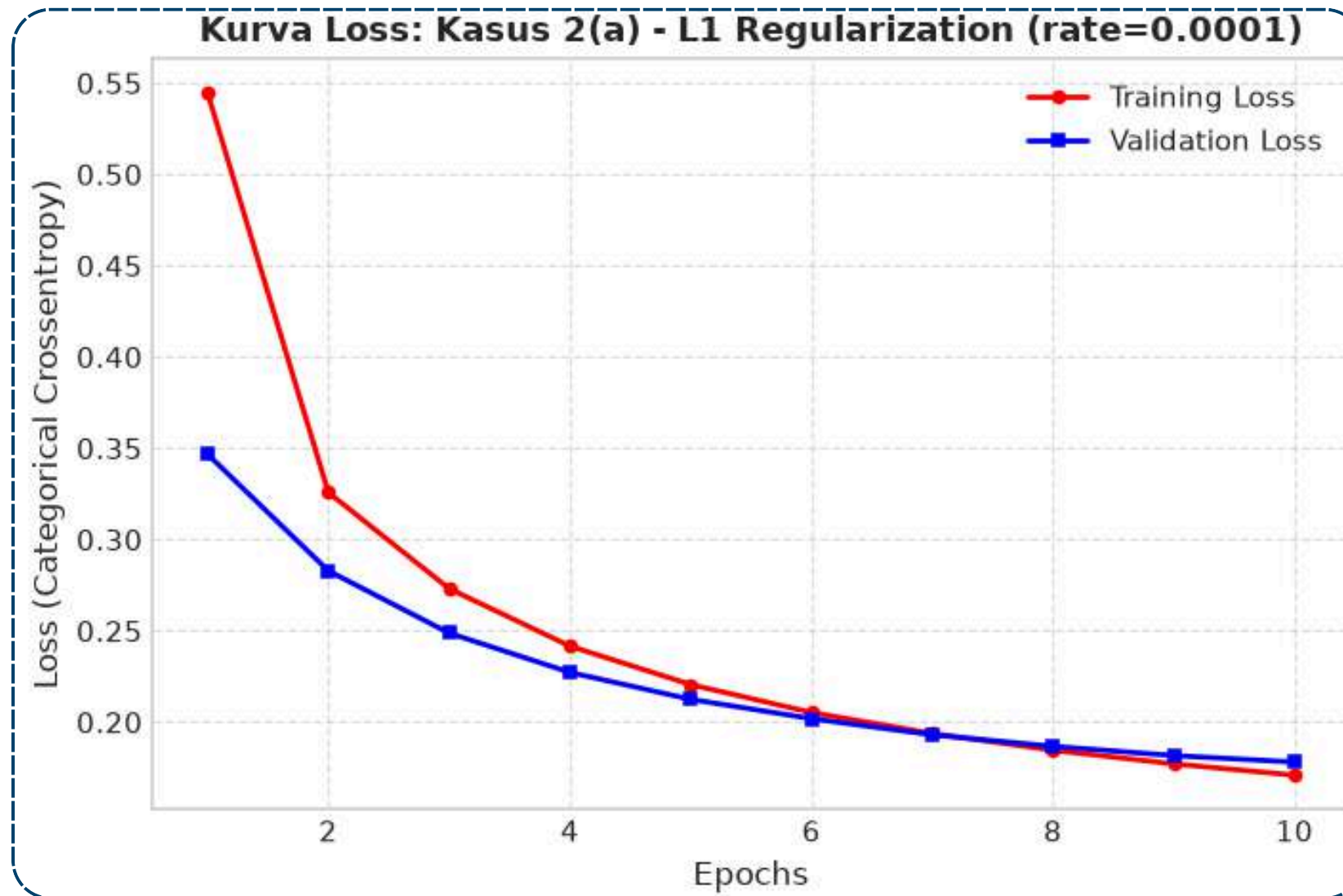
Tugas 2 Pembelajaran Mesin Mendalam

Oleh: DHIMAS PUTRA SULISTIO (24/537952/PA/22811)

Source URL

<https://github.com/muddglobb/DEEP-LEARNING/blob/main/Assignment%202/main.ipynb>

Hasil Skenario 2(a) - L1 Regularization



$$\lambda = 0.0001$$

Model Summary

50.890 Parameter

Flatten

0

Dense 64

50.240

Dense 10

650

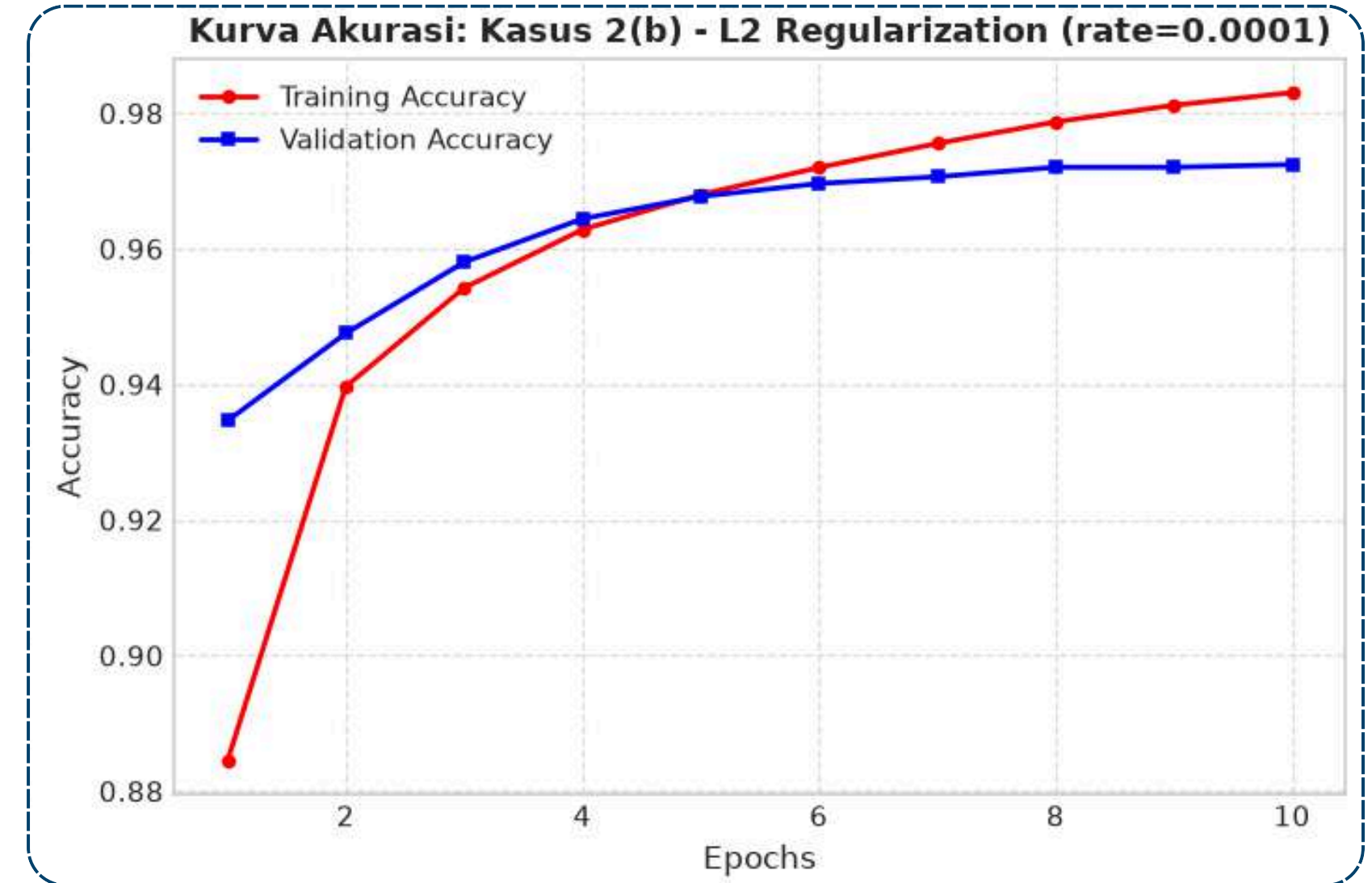
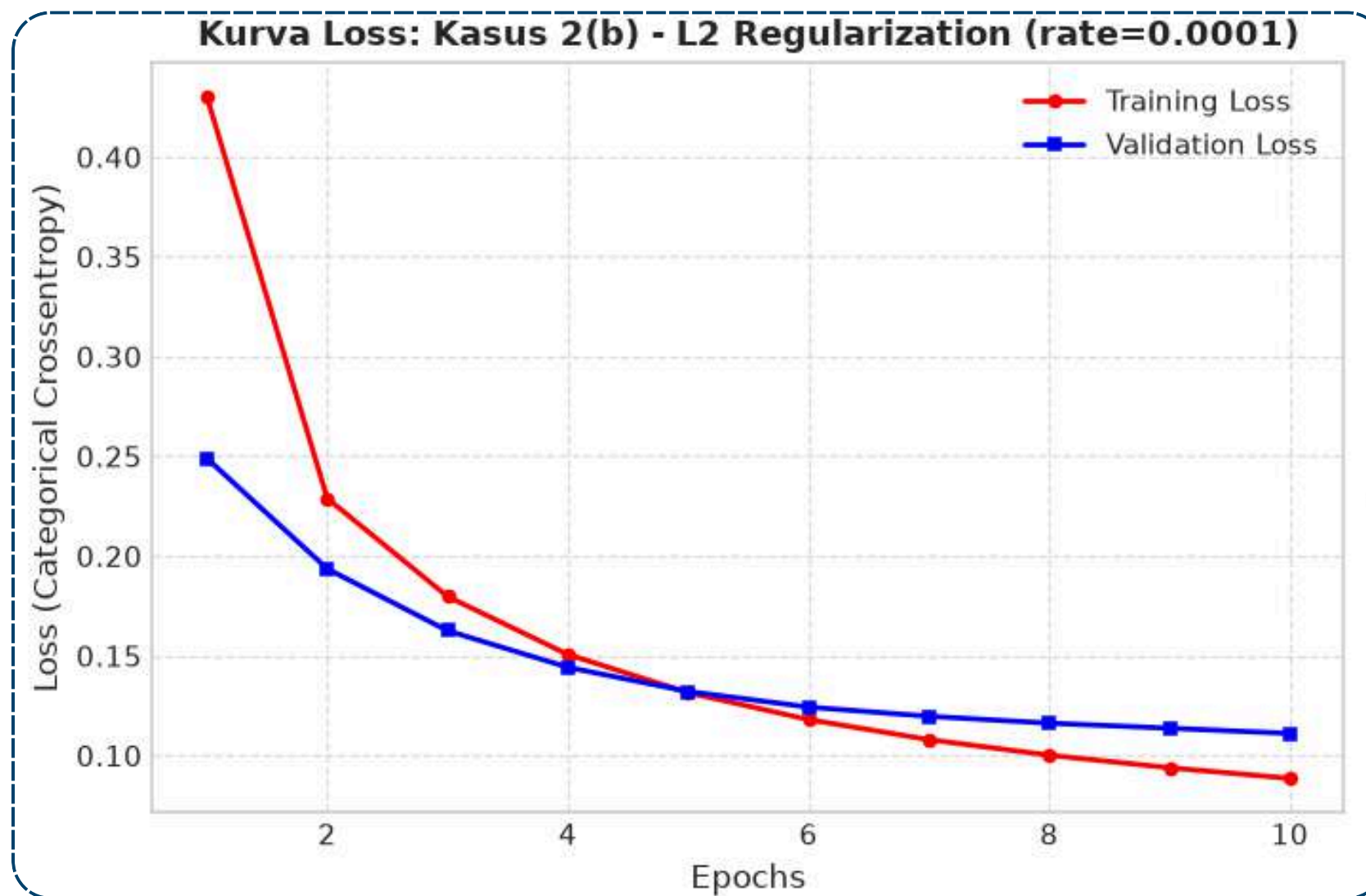
Test Loss

Nilai loss stabil pada data uji

Test Accuracy

~97%

Hasil Skenario 2(b) - L2 Regularization



$$\lambda = 0.0001$$

Model Summary

50.890 Parameter

Flatten

0

Dense 64

50.240

Dense 10

650

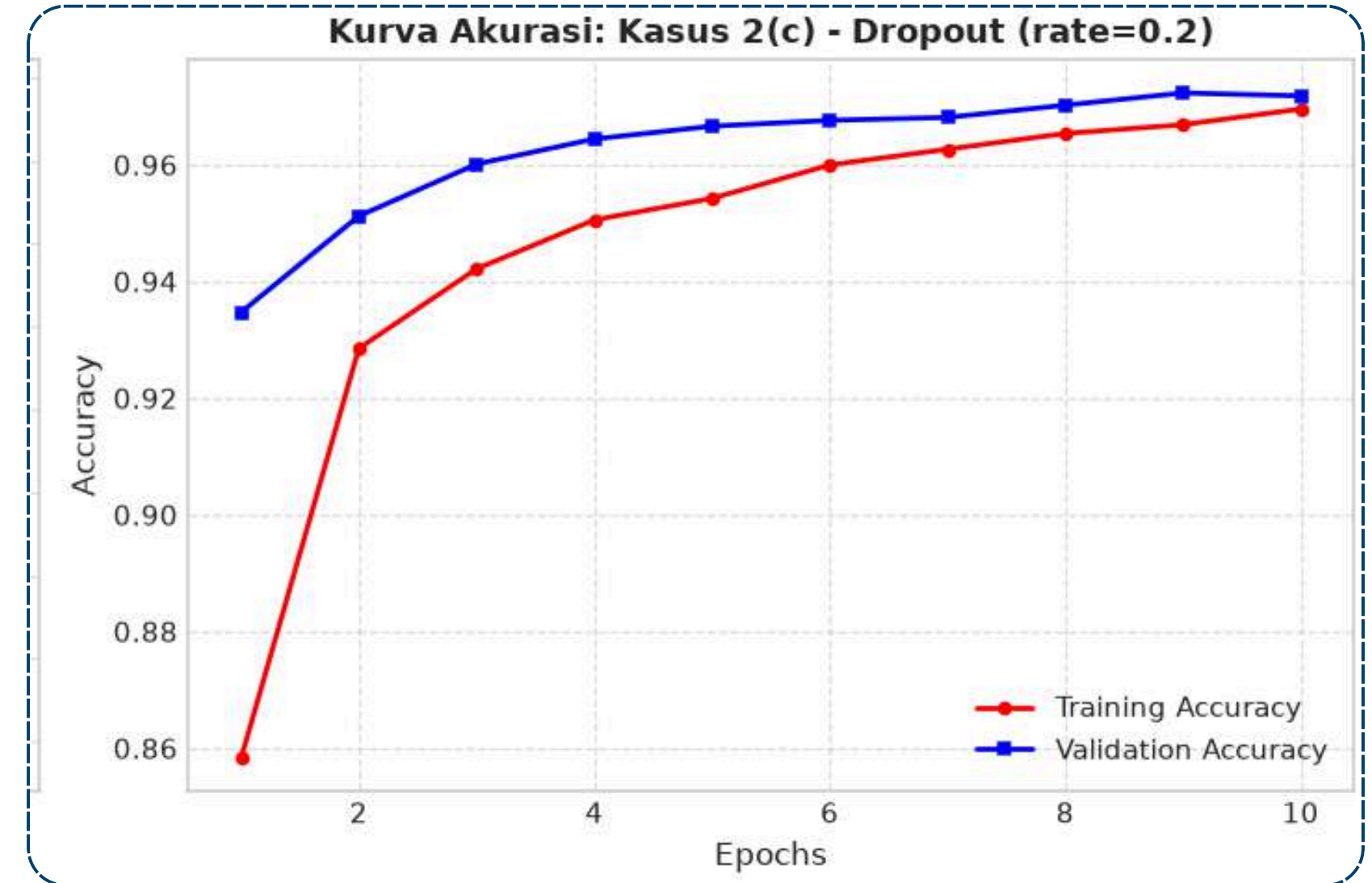
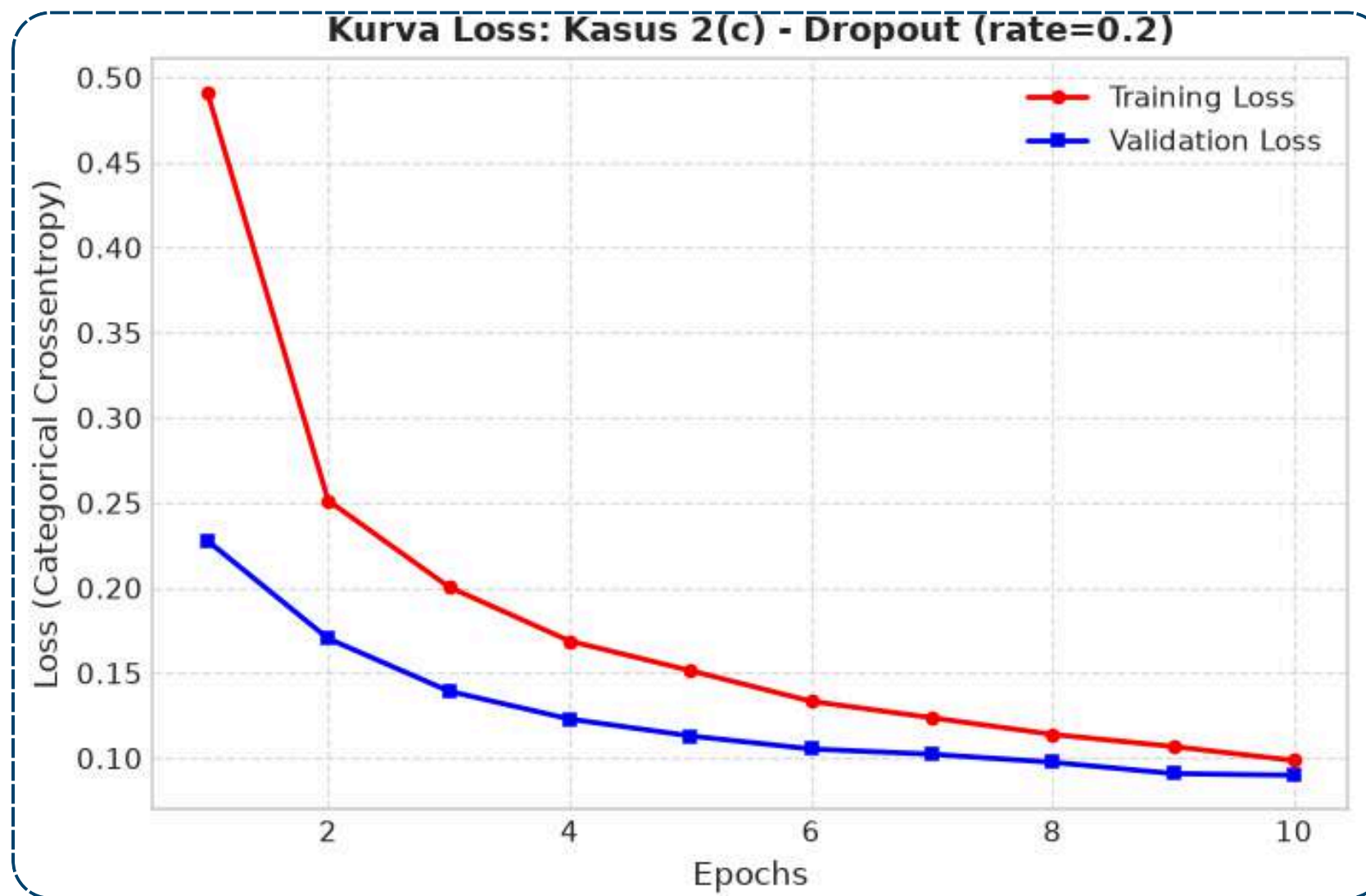
Test Loss

Sangat stabil
dengan kurva
smooth

Test Accuracy

~97.4%

Hasil Skenario 2(c) - Dropout



Dropout Rate
= 0.2

Model Summary

50.890 Parameter

Flatten

0

Dense 64

50.240

Dense 10

650

Drop Out

0

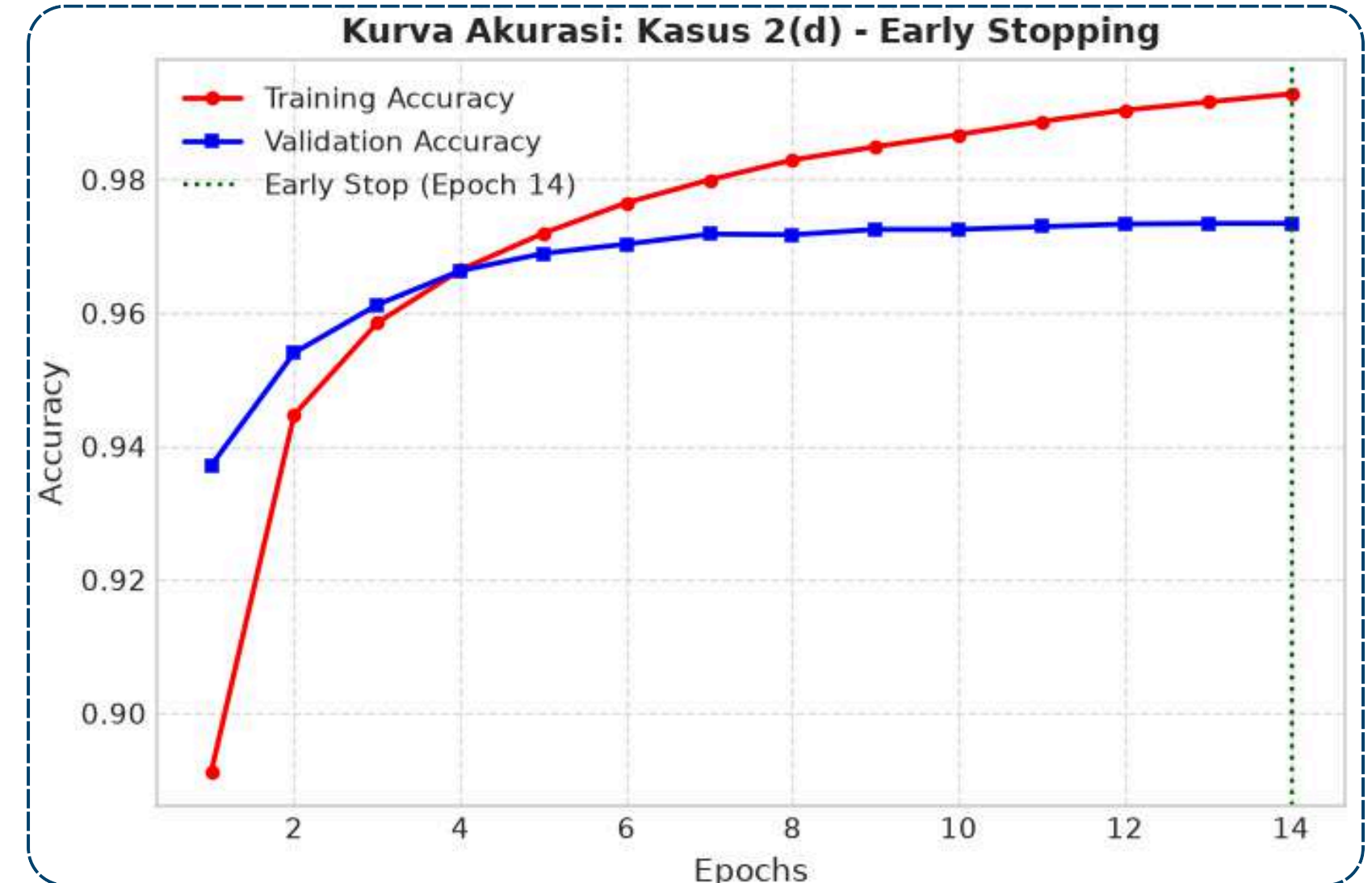
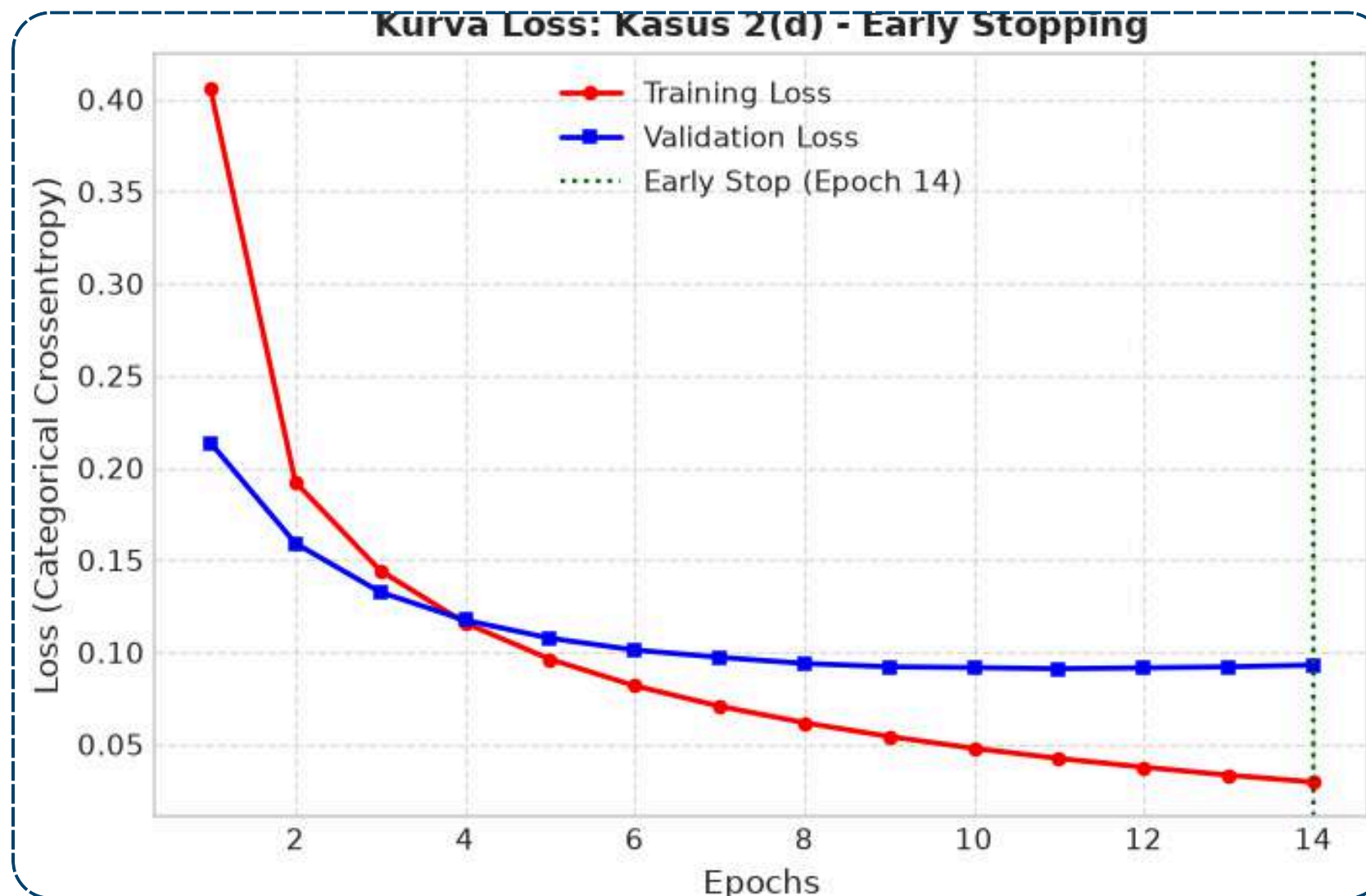
**Test
Loss**

~0.08

**Test
Accuracy**

~97.5%

Hasil Skenario 2(d) - Early Stopping



Monitor = 'val_loss'
Patience = 3
Restore Best Weight
= True

Model Summary

50.890 Parameter

Flatten
0

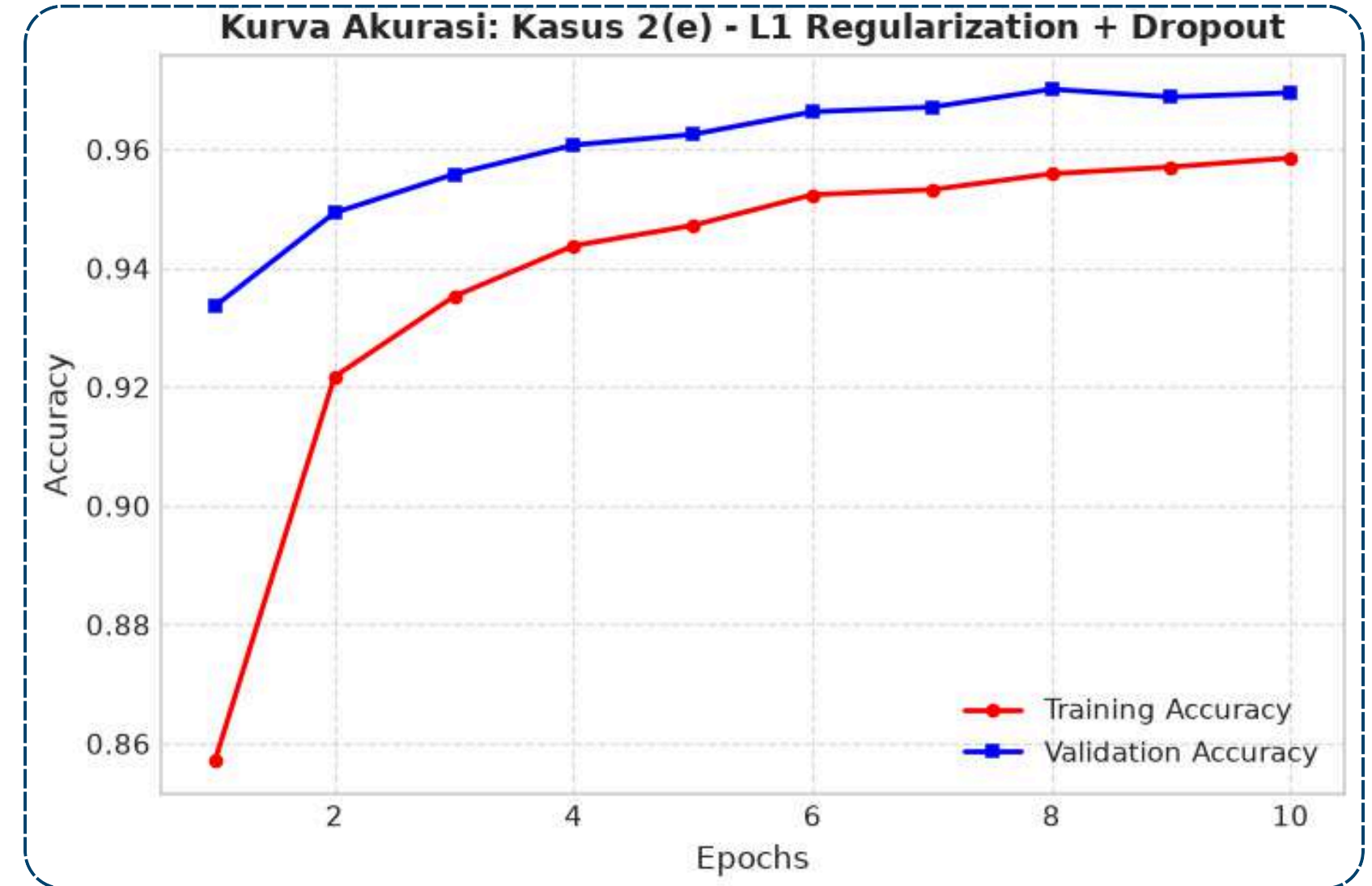
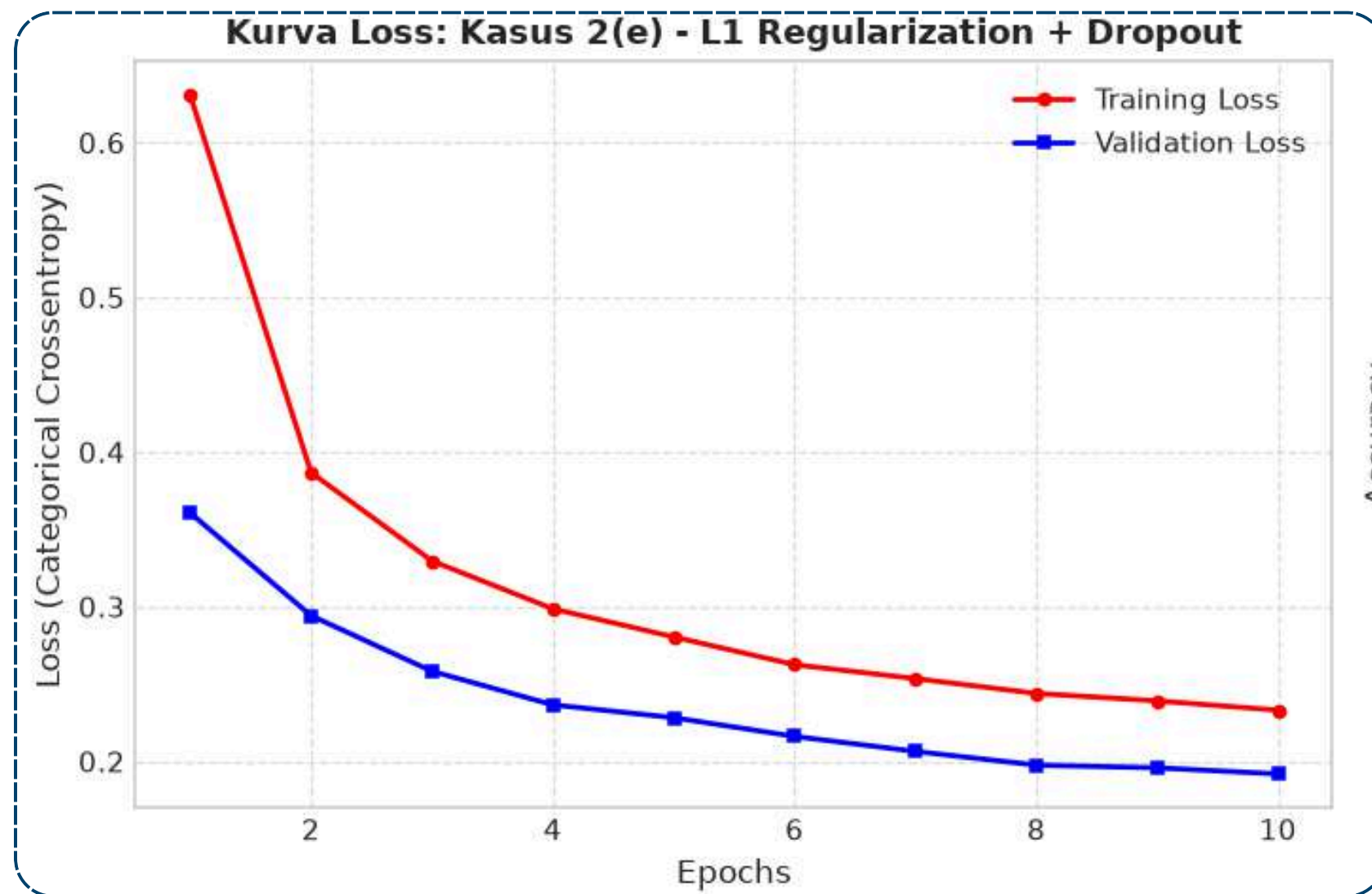
Dense 64
50.240

Dense 10
650

Pelatihan berhenti
sebelum epoch
maksimal ketika
validation loss
mulai jenuh.

**Test
Accuracy**
~97.5%
Bobot optimal
berhasil
dipulihkan.

Hasil Skenario 2(e) - L1 Regularization + Dropout



$\lambda = 0.0001$
Dropout Rate = 0.2

Model Summary

50.890 Parameter

Flatten

0

Dense 64

50.240

Dense 10

650

Drop Out

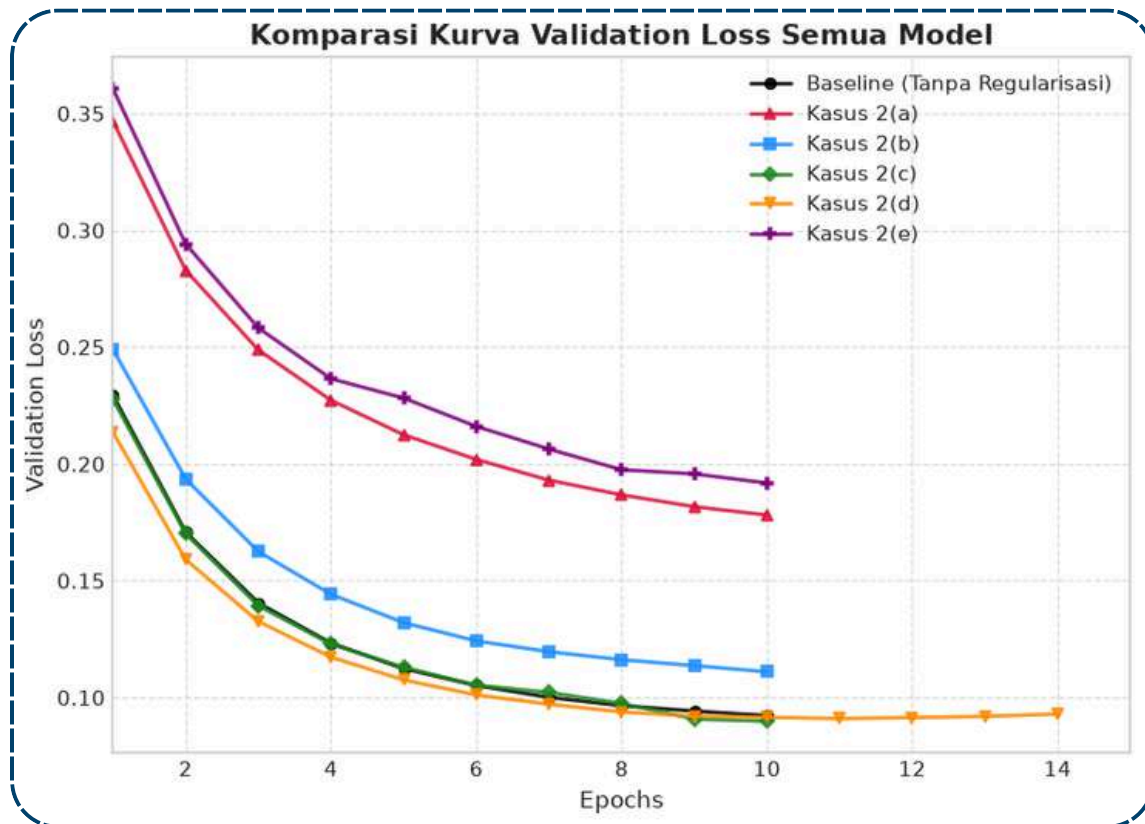
0

Memberi
kan
regulasi
ganda

Test
Accuracy

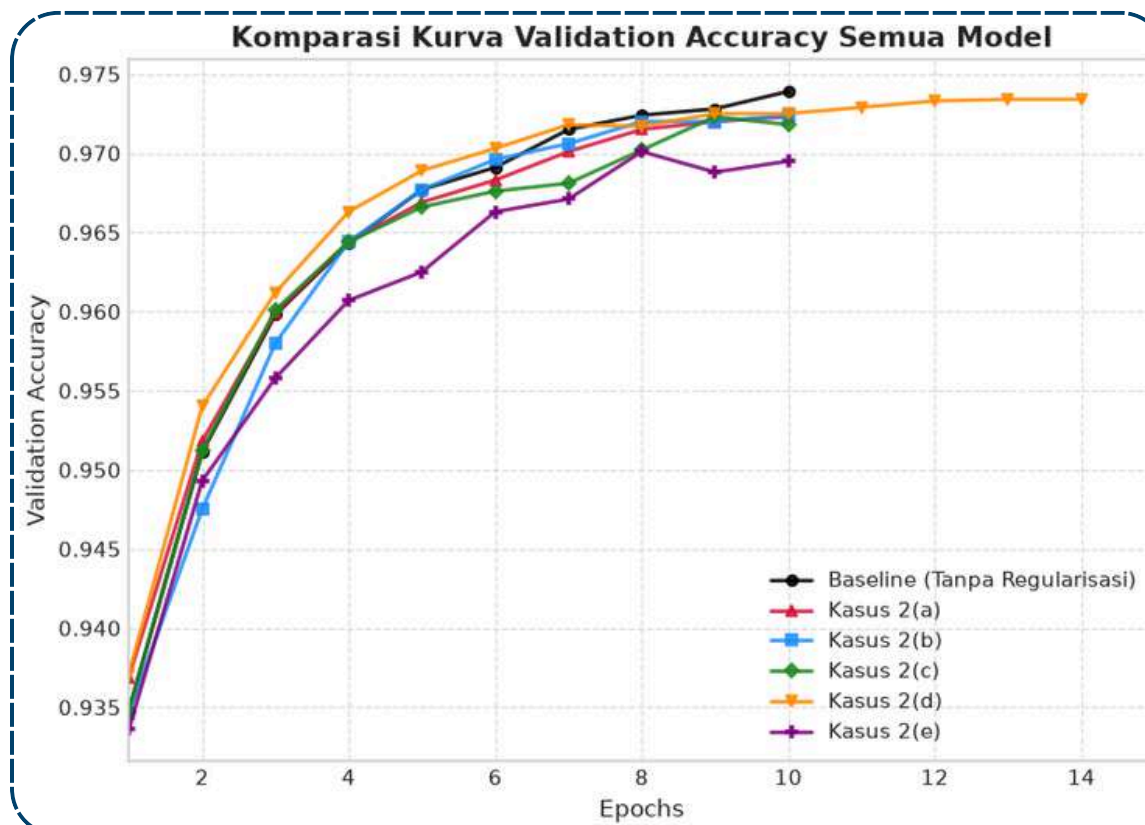
~97.2%

Kesimpulan



Baseline: Mencapai akurasi tinggi namun memiliki kecenderungan divergensi antara kurva training loss dan validation loss seiring bertambahnya epoch.

- **Dropout** dan **Early Stopping** memberikan keseimbangan paling optimal antara akurasi tinggi dan pencegahan overfitting.
- **L2 Regularization** menghasilkan kestabilan numerik dan kurva pelatihan yang paling smooth
- **L1 Regularization** terbukti secara empiris mendorong sparsitas representasi fitur



Untuk arsitektur jaringan saraf sederhana seperti klasifikasi MNIST, kombinasi **Dropout** (20%) bersama **Early Stopping** merupakan kombinasi paling efisien dan hasil terbaik.



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Terima Kasih.

*Merakyat, Mandiri, Berkelanjutan
Inclusive, Self-Reliant, Sustainable*

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